

Source Attribution and Willingness to Pay for Air Quality Forecasting in Lahore, Pakistan

A randomized controlled trial examining how attributing identical air quality forecasts to a government agency (EPD) vs. a citizen initiative (PAQI) affects consumer willingness to pay — and who responds most strongly.

Background

Lahore, Pakistan experiences severe air pollution with well-documented negative consequences for public health and educational outcomes. Two air quality forecasting services operate in this context: the **EPD** (Environmental Protection Department, government-run) and **PAQI** (Pakistan Air Quality Initiative, citizen-run).

This study asks: does the *source* to which an air quality forecast is attributed, holding forecast content constant, causally affect households’ willingness to pay (WTP) for that service? Both treatment arms received **identical** forecasting content; only the attributed source differed (EPD vs. PAQI).

Study Design

Parameter	Details
Setting	Shalamar & City Center Tehsils, Lahore
Design	2-arm RCT (EPD vs. PAQI attribution)
Sample	929 households after cleaning
EPD group	467 households
PAQI group	462 households
WTP elicitation	BDM (Becker–DeGroot–Marschak)
Survey	Two-stage: baseline + endline

Three outcomes were constructed:

- WTP for PAQI alone
- WTP for EPD alone

Relative WTP = $WTP_{PAQI} - WTP_{EPD}$ (primary outcome; SD = 24.7 PKR)

Absolute WTP for both services had nearly identical distributions ($\bar{x} \approx 228\text{--}229$ PKR, SD ≈ 68.7 PKR), consistent with treatment shifting relative preferences without changing overall spending.

Methods

The analysis proceeds in two stages with distinct objectives.

Stage 1 — What predicts WTP?

Three methods to identify robust predictors across all outcomes (WTP_{PAQI} , WTP_{EPD} , relative WTP):

- OLS** with BH correction across 6 model combinations (2 groups $\times$ 3 outcomes).
- Lasso** with 10-fold CV and a 70/30 train-test split.
- Bayesian Spike-and-Slab** (BoomSpikeSlab) for probabilistic variable selection.

All methods converge: treatment is the sole robust predictor of relative WTP.

Stage 2 — Who responds most strongly?

A **Causal Forest** (GRF; 4,000 trees; honest splitting; 50+ predictors) was used.

All OLS models use **cluster-robust standard errors** (clustered by grid, n=101 clusters).

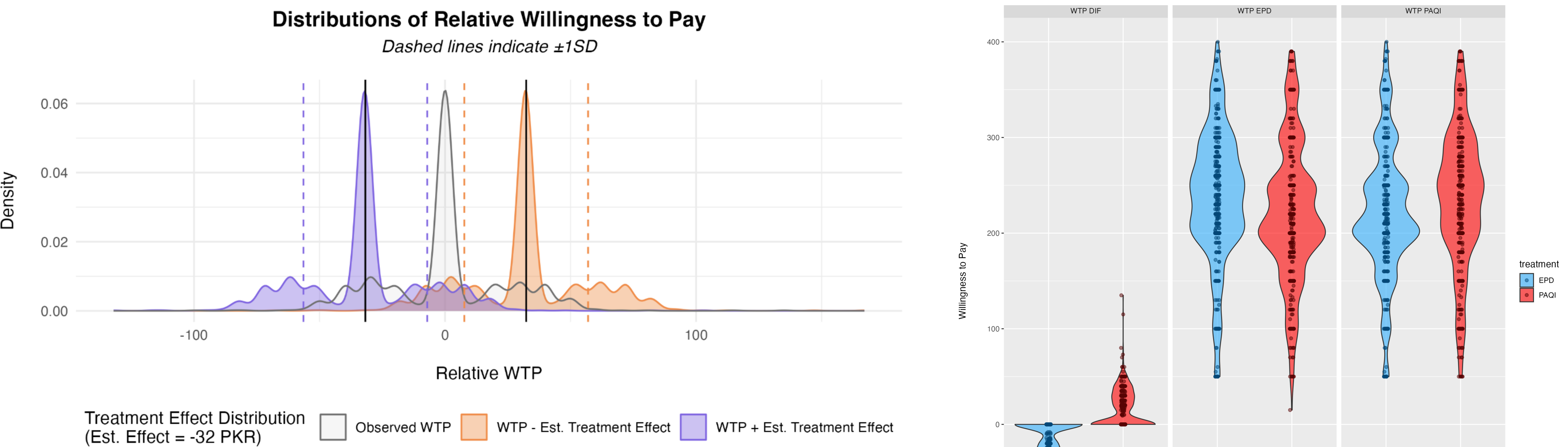
Main Result: Treatment Effect on Relative WTP

Cohen’s $d = 1.71$
95% CI: [1.56, 1.86] · Conventional “large” threshold: $d = 0.8$

Households assigned to EPD-attributed forecasting shift their relative WTP by approximately -32 PKR on average — larger than one full standard deviation of the outcome distribution (SD = 24.7 PKR). The effect is more than twice the conventional threshold for a large effect.

Quantity	Value
Treatment coefficient (relative WTP)	-32.0 PKR
Cohen’s d	1.71
95% CI for d	[1.56, 1.86]

With forecast content held constant, source attribution is the *only* available cue. Respondents lacking well-formed prior preferences anchor almost entirely to the attributed source. Logistic regression confirms near-perfect separation on the binary preference outcome: EPD-treated households show $\approx 97\%$ probability of preferring EPD; PAQI-treated households show $\approx 5\%$.



Left: Distribution of relative WTP by treatment group, with ± 1 SD bands. Right: Absolute WTP for EPD and PAQI across treatment arms — distributions are nearly identical, confirming treatment shifts relative preferences without changing total demand.

Key Findings

1. Source attribution has an unusually large causal effect. Cohen’s $d = 1.71$ (CI: [1.56, 1.86]); treatment explains $\sim 6\times$ more variance in relative WTP than all demographic, behavioral, and household variables combined. Direction was expected, dominance was not.

2. No covariates robustly predict WTP. Demographic and household variables fail to predict absolute WTP across methods. This is a positive convergent result, not merely a null and motivates the shift to causal forest heterogeneity analysis.

3. Effect operates through relative preferences, not total demand. Treatment leaves total WTP unaffected ($R^2 \approx 0$, $p = 0.831$), isolating the mechanism as preference formation between services rather than demand creation.

4. Income and location moderate treatment response. Work hours explains $\sim 36\%$ of treatment effect heterogeneity. Higher earners and Shalamar households respond more strongly; prior government approval further moderates response.

5. Policy implication. Targeted communication campaigns could shift preferences at low cost. Real-world effect magnitudes would likely be smaller, given the unusually salient cue in the survey environment.

Future work should examine the determinants of absolute WTP for air quality forecasting in Lahore. Baseline WTP of 229 PKR (9% of daily wages) suggests overall demand may be low independent of source attribution, and interventions targeting awareness of health consequences may be more impactful than branding campaigns alone.

Teo Richard
University of California, Davis
Undergraduate, Economics and Statistics
Advisor: Prof. Arman Rezaee
February 2026

Variance Decomposition

Model specification	R^2	p -value
Treatment only	0.424	< 0.001
All covariates (no treatment)	0.071	0.381 (n.s.)
Treatment + all covariates	0.460	< 0.001

Cluster-robust standard errors throughout. $n = 929$.

Treatment explains approximately **$6\times$ more variance** in relative WTP than every other measured variable combined. Adding all covariates increases R^2 by only 3.6 percentage points.

Treatment explains essentially zero variance in total WTP ($R^2 \approx 0$, $p = 0.831$). The mechanism is preference formation between services, not demand creation.

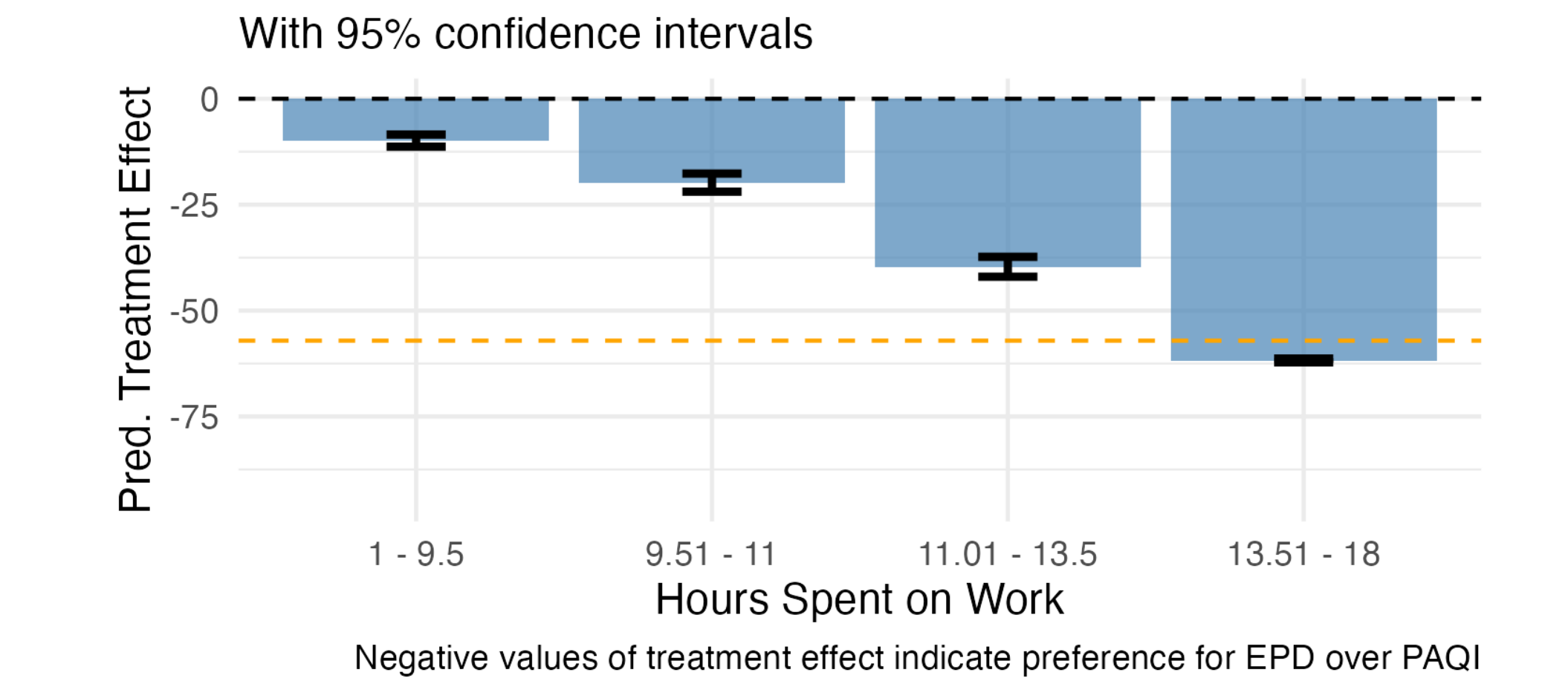
Treatment Effect Heterogeneity

Having established treatment as the dominant predictor, causal forest analysis characterizes *who responds most strongly*. Three variables together explain **$\sim 54\text{--}58\%$** of treatment effect variation:

Variable	Full	Trimmed	Interpretation
Work hours (total)	$\sim 35\%$	$\sim 36\%$	Income proxy; dominant moderator
Tehsil (location)	$\sim 12\%$	$\sim 13\%$	Shalamar > City Center effect
Govt. approval	$\sim 8\%$	$\sim 5\%$	Prior attitudes moderate response

“Trimmed” = work hours at 5th–95th percentile ($n = 876$). Rankings consistent across both analyses.

Work hours is the dominant moderator: each additional work hour per week decreases the treatment effect by **$8.2\text{--}9.8$ PKR** (SE ≈ 0.4 , $p < 0.001$), consistent with evidence that lower-income households in Pakistan hold lower trust in government institutions.



Limitations

Exploratory design. All analyses are exploratory and data-driven. Results should be treated as hypothesis-generating rather than confirmatory.

Stated vs. revealed preferences. Source attribution being the only available signal likely overstates the effect magnitude relative to naturalistic settings.

Heterogeneity estimates are exploratory. Causal forest variable importance percentages should be interpreted directionally.